

Cboe Titanium Cboe Japan Equities Binary Order Entry Version 3 Specification

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1. Introduction

1.1 Overview

This document describes Binary Order Entry, version 3 (BOEv3), the Cboe Japan proprietary order entry protocol used by participants to send orders to Cboe Japan.

Where applicable, the terminology (e.g., time in force) used in this document is similar to that used by the FIX protocol to allow those familiar with FIX to more easily understand BOEv3. This document assumes the reader has basic knowledge of the FIX protocol.

BOEv3 fulfills the following requirements:

- *CPU and memory efficiency.* Message encoding, decoding, and parsing are simpler to code and can be optimized to use less CPU and memory at runtime.
- *Application level simplicity.* State transitions are simple and unambiguous. They are easy to apply to a participant's representation of an order.
- *Session level simplicity.* The session level protocol (login, sequencing, replay of missed messages, logout) is simple to understand.

While Cboe Japan strives to preserve feature parity between FIX and BOEv3 where possible, some features may only be available in one protocol or the other.

Note that while FIX is an ASCII based protocol, BOE is binary based, providing for efficiencies that can allow for reduced latency.

All binary values are in **little-endian** (as used by Intel x86 processors), and not network byte order.

Each message is identified by a unique message type. A listing of the supported message types is provided in Table 1 - Session Message Types and Table 2 - Application Message Types.

All communication is via standard TCP/IP.

1.2 BOEv3 Message Format Versioning and Nomenclature

Message types may be introduced when new fields cannot be accommodated by utilizing reserve bytes in the existing message specification. In such cases, when a new message type is introduced, it will be documented as a distinct message type in this document. Application layer message types are named using the following pattern:

Application Layer Message Type	General Type	Market	Version
NewOrderJPEquitiesV1	New Order	JP Equities	V1

This allows for ease of distinction between similar message types between markets (for example, JP Equities compared to US Futures), and the handling of new versions of the message (V1, V2, *et cetera*).

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When application layer message types are discussed in this document, they are referred to using their general type name unless their specific version is relevant to the documentation.

During any time when multiple versions of messages from Cboe Japan to Participant are supported concurrently, the configuration of the port determines which message version may be sent by Cboe Japan.

1.2.1 Introduction of New Fields in Existing BOEv3 Messages

Existing message fields will not change in length. Fields currently identified as "Reserved" may be redefined, in part or in whole, as new specified fields. New fields may be introduced at the end of any message that does not have a repeating group of fields; consequently, Clients must check the length of messages received from Cboe Japan and treat any additional bytes present as undefined values.

1.3 Hours of Operation

Refer to the website for the Cboe Japan trading hours and Cboe Japan holiday schedule.

1.4 Data Types

The following data types are used by BOEv3. The size of some data types varies by message. All data types have default values of binary zero, in both Participant to Cboe Japan and Cboe Japan to Participant contexts.

- *Binary:* Little Endian byte order, unsigned binary value. The number of bytes used depends on the context.
    - One byte: FE = 254
    - Four bytes: 64 00 00 00 = 100
- *Binary Price:* Little Endian byte order value, signed two's complement, eight bytes in size, with four implied decimal places. So, if the value is 123,400, the actual value considering implied decimal places is 12.34.
    - 08 E2 01 00 00 00 00 00 = 123,400/10,000 = 12.34
    - F8 1D FE FF FF FF FF FF = -123,400/10,000 = -12.34
- *Alpha:* ASCII uppercase letters (A-Z) and lowercase letters (a-z) only. ASCII NUL (0x00) filled on the right, if necessary. The number of bytes used depends on the context.
- *Alphanumeric:* ASCII uppercase letters (A-Z), lowercase letters (a-z) and numbers (0-9) only. ASCII NUL (0x00) filled on the right, if necessary.
- *Text:* Printable ASCII characters only (binary values in the inclusive range 0x20 through 0x7E). ASCII NUL (0x00) filled on the right, if necessary. The number of bytes used depends on the context.
- *DateTime:* Little Endian byte order, unsigned binary value, 8 bytes in size. The date and time, in UTC, represented as nanoseconds past the UNIX epoch (00:00:00 UTC on 1 January 1970). The nanoseconds portion is set to 0 by the exchange (the actual precision will be in microseconds).
    - E0 FE 20 F7 36 71 F8 11 = 1,294,909,373,757,325,000 = 2011-01-13 09:02:53.757325 UTC

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- *Date:* Little Endian byte order, unsigned binary value, 4 bytes in size. The YYYYMMDD expressed as an integer.
    - A7 3C 34 01 = 20200615 = "June 15, 2020"
- *Reserved*: sequence of ASCII NUL (0x00) values when sent by the participant. May contain any values when sent by the exchange and should be ignored by the participant.

1.5 Protocol Features

The exchange does not guarantee messages sent by Members to the exchange, including through protocols such as TCP. Members are responsible to monitor the status of the messages they send to the exchange.

1.5.1 Architecture

Each BOE order handler process will allow a single TCP connection from a participant. Connection attempts from unknown source IP ranges will be blocked to prevent unauthorized access to BOE ports. Contact the Cboe Japan Trade Desk to connect from a new source IP range.

Each BOE order handler will connect, using a proprietary UDP protocol, to all matching units.

1.5.2 Alpha Segregation from Select

The Select market is deployed independently of the Alpha market. Consequently, a BOE port must be ordered specific to either the Select markets or the Alpha market.

1.5.3 Maximum Open Order Limits

The exchange limits the maximum number of open orders allowed on a BOE port to 300,000 per port. New orders will be rejected once this limit is breached until the number of open orders drops back below the limit.

1.5.4 Port Types

All BOE port types may be ordered through the Customer Web Portal using the Logical Port Request tool. Changes to port attributes may be requested through the same tool by submitting a 'Modify' request for one or more existing BOE ports. Note that to interact with the Select market, a port must be ordered for that market.

1.5.4.1 BOE Order Ports

BOE Order Ports support order entry, modification, and cancellation. Order Ports purposefully do not support the usage of the `Purge Orders` message type. The attempted usage of this message type on a standard BOE order port will result in a rejection of the disallowed message type.

BOE Order Ports are limited to 4,000 inbound messages per second. Once the inbound limit is reached new orders are rejected, modifies are handled as cancels, and cancels are processed normally. BOE Order Ports are limited to 10 inbound messages per second on Cboe Japan Test Products.

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1.5.4.2 BOE Matching Units and Sequence Numbers

BOE Order Ports are not specific to a particular matching unit. Note however that message sequence numbers from Cboe Japan are issued per matching unit. Sequence numbers will be monotonically incremental values per *MatchingUnit*. Currently, Cboe Japan has 2 matching units for Alpha, and 2 matching units for Select. August 3rd 2025 onwards, Cboe Japan will have 4 matching units for Alpha, and 4 matching units for Select.

1.5.4.3 BOE Purge Ports

BOE Purge Ports support only one message type from participants: `Purge Orders`. Participants may use this port type to request a cancellation of groups of orders, including orders across multiple BOE/FIX Order Ports.

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2. Session

2.1 Message Headers

Each message has a twelve-byte header. The two initial *StartOfMessage* bytes are present to aid in message reassembly for network capture purposes. The *MatchingUnit* field is only populated on sequenced, non-session level messages sent from Cboe Japan to the Participant. Messages from Participant to Cboe Japan and all session level messages must always set this value to 0.

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	`B0 E3` (58288)
MessageLength	2	2	Binary	Number of bytes for the message, including this field but not including the two bytes of the *StartOfMessage* field.
MessageType	4	2	Binary	Message type
MatchingUnit	6	1	Binary	Matching unit which created this message. Matching units in BOE correspond to matching units on Multicast PITCH. For session level traffic the unit is set to 0. For messages from Participant to Cboe Japan, the unit must be 0.
Reserved	7	1	Binary	Must be zero from participant. Value unspecified from Cboe Japan.
SequenceNumber	8	4	Binary	The sequence number for this message. Messages from Cboe Japan to Participant are sequenced distinctly per matching unit. Zero for session level traffic.

2.1.1 Handling of Invalid Message Headers

If an invalid message header is encountered, the Exchange will disconnect the port. A message header is considered invalid if any of the following is encountered:

- *StartOfMessage* is not `B0E3`.
- *MessageLength* is not appropriate for the given message type.
- *MessageType* is not a documented message type for Cboe Japan. Note that the types of application messages accepted will vary between Cboe Japan and other BOEv3 exchanges.

2.2 Login, Replay, and Sequencing

Session level messages, both inbound (Participant to Cboe Japan) and outbound (Cboe Japan to Participant) are unsequenced.

Inbound (participant to Cboe Japan) application messages are sequenced. Upon reconnection, Cboe Japan informs the participant of the last processed sequence number; the participant may choose to resend any messages with sequence numbers greater than this value. A gap forward in the participant's

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incoming sequence number is permitted at any time and is ignored by Cboe Japan. Gaps backward in sequence number (including the same sequence number used twice) are never permitted and will always result in a `Logout Response` message being sent and the connection being dropped.

Most (but not all) outbound (Cboe Japan to participant) application messages are monotonically sequenced per matching unit. Each message's documentation will indicate whether it is sequenced or unsequenced. While matching units on BOE correspond directly to matching units on Multicast PITCH, sequence numbers do not.

Upon reconnection, a participant sends the last received sequence number per matching unit in a `Login Request` message. Cboe Japan will then respond with any missed messages.

The *ReplayUnspecifiedUnit* value can be used to control the replay behavior for unknown units. If the flag is set to F (Fail), Cboe Japan will send a `Login Response` and close the connection if the login request does not list all matching units. If the flag is set to S (Skip), Cboe Japan will exclude messages from unspecified matching units during replay. If the flag is set to R (Replay), Cboe Japan will send messages from unspecified units during replay (starting with sequence 1 per unspecified unit). Cboe Japan will send a `Replay Complete` message when replay is finished. If there are no messages to replay, a `Replay Complete` message will be sent immediately after a `Login Response` message. Cboe Japan will reject all orders during replay.

If a participant has requested replay messages using a properly formatted `Login Request` after a disconnect, any unacknowledged orders remaining with the participant after the `Replay Complete` message is received should be assumed to be unknown to Cboe Japan.

Unsequenced messages will not be included during replay.

A session is identified by the *SessionId* and *SessionSubId* (both supplied by Cboe Japan).

If a login is rejected an appropriate `Login Response` message will be sent and the connection will be terminated.

See 'Appendix A – Login Strategies' for recommended login messaging.

2.3 Sequence Reset

A reset sequence operation is not available for Binary Order Entry. However, a participant can send a `Login Request` message with *ReplayUnspecifiedUnit* set to S (Skip), and *NumberOfUnits* set to zero. Then, upon receiving a `Login Response` message from Cboe Japan, the participant can use the *ClientSequence* value in that message as the sequence starting point for sending future messages.

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2.4 Heartbeats

`Client Heartbeat` messages are sent from participant to Cboe Japan, and `Server Heartbeat` messages are sent from Cboe Japan to Participant if no other data has been sent in that direction for one second. Similar to other session level messages, heartbeats from Cboe Japan to the participant do not increment the sequence number. If Cboe Japan receives no inbound data or heartbeats for five seconds, a `Logout Response` message will be sent, and the connection will be terminated. **Clients are encouraged to have a one second heartbeat interval and to perform similar connection staleness logic.**

2.5 Logging Out

To gracefully log out of a session, a `Logout Request` message should be sent by the participant. Cboe Japan will finish sending any queued data for that port, respond with a `Logout Response` message, and close the connection. After receipt of a `Logout Request` message, Cboe Japan will ignore all other inbound (Participant to Cboe Japan) messages except for `Client Heartbeat`.

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3. Session Messages

Table 1 - Session Message Types

Direction	Message Name	Type	Sequenced
Participant to Cboe	Login Request	`01 00` (1)	No
Participant to Cboe	Logout Request	02 00 (2)	No
Participant to Cboe	Client Heartbeat	`03 00` (3)	No
Cboe to Participant	Login Response	`F5 01` (501)	No
Cboe to Participant	Replay Complete	`F6 01` (502)	No
Cboe to Participant	Logout Response	`F7 01` (503)	No
Cboe to Participant	Server Heartbeat	`F8 01` (504)	No

3.1 Participant to Cboe Japan

3.1.1 Login Request

A `Login Request` message must be sent as the first message upon connection.

This message includes a repeating group starting with field *UnitNumber* which repeats *NumberOfUnits* times. *NumberOfUnits* can be used to specify the last consumed sequence numbers per matching unit received by the participant. Cboe Japan uses these sequence numbers to determine what outbound (Cboe Japan to participant) traffic, if any, was missed by the participant. If *NumberOfUnits* is 0, it is assumed the participant has not received any messages (e.g., start of day).

The participant does not need to include a sequence number for a unit if they never received messages from it. If the participant wishes to send a value for the unit anyway, 0 is the only allowed value.

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	`B0 E3` (58288)
MessageLength	2	2	Binary	(30 + 5***NumberOfUnits*)
MessageType	4	2	Binary	`01 00` (1)
MatchingUnit	6	1	Binary	Must be zero
Reserved	7	1	Binary	Must be zero
SequenceNumber	8	4	Binary	Must be zero
SessionId	12	4	AlphaNumeric	Session Id as supplied by Cboe Japan
SessionSubId	16	4	AlphaNumeric	Session Sub Id as supplied by Cboe Japan (ex. “0001”)
Password	20	10	AlphaNumeric	The password associated with the *SessionId* and *SessionSubId*.

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ReplayUnspecifiedUnit	30	1	Text	Controls replay behavior for unknown units. Must be one of: F = fail if any unit is not specified R = replay any unspecified unit from one S = skip replay of unspecified units
NumberOfUnits	31	1	Binary	The number (possibly 0) of unit/sequence pairs to follow, one per unit from which the participant has received messages over this port. This value must be 0, 1, or 2 since Cboe Japan has 2 matching units. August 3rd 2025 onwards, the number (possibly 0) of unit/sequence pairs to follow, one per unit from which the participant has received messages over this port. This value must be 0, 1, 2, 3, or 4 since Cboe Japan has 4 matching units.
→*UnitNumber*	32	1	Binary	A unit number (1 , 2, since Cboe Japan has 2 matching units) August 3rd 2025 onwards, a unit number (1, 2, 3, 4 since Cboe Japan has 4 matching units).
→*UnitSequence*	33	4	Binary	Last received sequence number for the unit (Zero if nothing has been received).

3.1.1.1 Example Login Request Message

Field Name	**Hexadecimal**	**Description**
StartOfMessage	B0 E3	
MessageLength	28 00	40
MessageType	01 00	1 (login request)
MatchingUnit	00	
Reserved	00	
SequenceNumber	00 00 00 00	(unsequenced)
SessionId	41 42 43 44	(ex. “ABCD”) - Session Id as supplied by Cboe Japan
SessionSubId	30 30 30 31	(ex. “0001”) Session Sub Id as supplied by Cboe Japan
Password	61 62 63 64 65 66 67 68 69 6A	(ex. “abcdefghij”) Password
ReplayUnspecifiedUnit	52	“R” - Replay unknown units.
NumberOfUnits	02	2 units
→UnitNumber	01	Unit 1

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→UnitSequence	00 00 00 00	0
→UnitNumber	02	Unit 2
→UnitSequence	00 00 00 00	0
From 3rd August 2025 onwards:		
NumberOfUnits	04	4 units
→UnitNumber	01	Unit 1
→UnitSequence	00 00 00 00	0
→UnitNumber	02	Unit 2
→UnitSequence	00 00 00 00	0
→UnitNumber	03	Unit 3
→UnitSequence	00 00 00 00	0
→UnitNumber	04	Unit 4
→UnitSequence	00 00 00 00	0

3.1.2 Logout Request

To end the session, the participant should send a `Logout Request` message. Cboe Japan will finish sending any queued data, respond with a `Logout Response` message, and close the connection.

A participant may simply close the connection without logging out however, queued messages may be lost when closing the connection without logging out.

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	`B0 E3` (58288)
MessageLength	2	2	Binary	`0A 00` (10)
MessageType	4	2	Binary	`02 00` (2)
MatchingUnit	6	1	Binary	Must be zero
Reserved	7	1	Binary	Must be zero
SequenceNumber	8	4	Binary	Must be zero

3.1.2.1 Example Logout Request Message

Field Name	Hexadecimal	Description
StartOfMessage	B0 E3	Start of message bytes
MessageLength	0A 00	10 bytes
MessageType	02 00	Logout Request
MatchingUnit	00	Must be zero
Reserved	00	Must be zero
SequenceNumber	00 00 00 00	Must be zero

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3.1.3 Client Heartbeat

See 'Section 2.4 - Heartbeats' for more information about heartbeats and the session level protocol.

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	`B0 E3` (58288)
MessageLength	2	2	Binary	`0A 00` (10)
MessageType	4	2	Binary	`03 00` (3)
MatchingUnit	6	1	Binary	Must be zero
Reserved	7	1	Binary	Must be zero
SequenceNumber	8	4	Binary	Must be zero

3.1.3.1 Example Client Heartbeat Message

Field Name	Hexadecimal	Description
StartOfMessage	B0 E3	Start of message bytes
MessageLength	0A 00	10 bytes
MessageType	03 00	Client Heartbeat
MatchingUnit	00	Must be zero
Reserved	00	Must be zero
SequenceNumber	00 00 00 00	Must be zero

3.2 Cboe Japan to Participant

3.2.1 Login Response

`A Login Response` message is sent in response to a `Login Request` message. On a successful login, the *LoginResponseStatus* will be set to 'A'. On a failed login, *LoginResponseStatus* will be set to a value other than 'A', and *LoginResponseText* will be set to an appropriate failure description.

Note that the repeating group starting with field *UnitNumber* provides the highest available Cboe Japan to participant sequence number for the unit. All units will be included in a successful `Login Response`, regardless of whether all units were listed in the `Login Request` message.

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	`B0 E3` (58288)
MessageLength	2	2	Binary	(76 + NumberOfUnits*5)
MessageType	4	2	Binary	`F5 01` (501)
MatchingUnit	6	1	Binary	Will be zero
Reserved	7	1	Binary	Unspecified
SequenceNumber	8	4	Binary	Will be zero
LoginResponseStatus	12	1	Text	Accepted, or the reason for the rejection.

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				A = Accepted B = Session in Use D = Disabled I = Invalid Unit M = Invalid Message (first message was not Login Request) N = Not Authorized Q = Sequence Ahead S = Invalid Session
LoginResponseText	13	60	Text	Human-readable text with additional information about the reason for rejection. ASCII NUL (0x00) filled on the right, if necessary.
ClientSequence	73	4	Binary	Last inbound (participant to Cboe Japan) message sequence number processed by Cboe Japan on this port.
NumberOfUnits	77	1	Binary	A number, n, of unit/sequence pairs to follow, one per unit. A pair for every unit will be sent, even if no messages have been sent to this port today. For unsuccessful logins, this will be 0.
→*UnitNumber*	78	1	Binary	A unit number
→*UnitSequence*	79	4	Binary	Highest available Cboe Japan to participant sequence number for the unit.

3.2.1.1 Example Login Response Message

Field Name	Hexadecimal	Description
StartOfMessage	B0 E3	Start of message bytes
MessageLength	56 00	86 bytes
MessageType	F5 01	Login Response
MatchingUnit	00	Always zero
Reserved	00	
SequenceNumber	00 00 00 00	Always zero
LoginResponseStatus	41	"A" (Accepted)
LoginResponseText	54 45 53 54 49 4E 47 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	"TESTING"
ClientSequence	01 00 00 00	1
NumberOfUnits	02	2 units

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→*UnitNumber*	01	Unit 1
→*UnitSequence*	06 41 27 00	2,572,550
→*UnitNumber*	02	Unit 2
→*UnitSequence*	3F 15 00 00	5,439

3.2.2 Replay Complete

See 'Section 2.2 - Login, Replay, and Sequencing' for more information about replay and the session level protocol.

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	`B0 E3` (58288)
MessageLength	2	2	Binary	`0A` (10)
MessageType	4	2	Binary	`F6 01` (502)
MatchingUnit	6	1	Binary	Will be zero
Reserved	7	1	Binary	Unspecified
SequenceNumber	8	4	Binary	Will be zero

3.2.2.1 Example Replay Complete Message

Field Name	Hexadecimal	Description
StartOfMessage	B0 E3	Start of message bytes
MessageLength	0A 00	10 bytes
MessageType	F6 01	Replay Complete
MatchingUnit	00	Always zero
Reserved	00	
SequenceNumber	00 00 00 00	Always zero

3.2.3 Logout Response

A `Logout Response` is usually sent in response to a `Logout Request`. Any queued data is transmitted, a `Logout Response` is sent, and Cboe Japan will close the connection. However, a `Logout Response` may also be sent if the participant violates the protocol specification (e.g., by moving backwards in sequence number).

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	`B0 E3` (58288)
MessageLength	2	2	Binary	`47 00` (71)
MessageType	4	2	Binary	`F7 01` (503)
MatchingUnit	6	1	Binary	Will be zero
Reserved	7	1	Binary	Unspecified

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SequenceNumber	8	4	Binary	Will be zero
LogoutReason	12	1	Text	Reason for the logout. U = User Requested A = Administrative ! = Protocol Violation (ex. timeout due to inactivity)
LogoutReasonText	13	60	Text	Human-readable text with additional information about the reason for logout.

3.2.3.1 Example Logout Response Message

Field Name	Hexadecimal	Description
StartOfMessage	B0 E3	Start of message bytes
MessageLength	47 00	71 bytes
MessageType	F7 01	Logout Response
MatchingUnit	00	Always zero
Reserved	00	
SequenceNumber	00 00 00 00	Always zero
LogoutReason	55	"U" (User Requested)
LogoutReasonText	54 45 53 54 49 4E 47 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	"TESTING"

3.2.4 Server Heartbeat

See 'Section 2.4 - Heartbeats' for more information about heartbeats and the session level protocol.

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	B0 E3 (58288)
MessageLength	2	2	Binary	0A (10)
MessageType	4	2	Binary	F8 01 (504)
MatchingUnit	6	1	Binary	Will be zero
Reserved	7	1	Binary	Unspecified
SequenceNumber	8	4	Binary	Will be zero

3.2.4.1 Example Server Heartbeat Message

Field Name	Hexadecimal	Description
StartOfMessage	B0 E3	Start of message bytes

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MessageLength	0A 00	10 bytes
MessageType	F8 01	Server Heartbeat
MatchingUnit	00	Always zero
Reserved	00	
SequenceNumber	00 00 00 00	Always zero

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4. Application Messages

Note that if Cboe Japan receives a message type not listed in this specification, the connection will be closed. See section "2.1.1 - Handling of Invalid Message Headers" for details.

Table 2 - Application Message Types

Direction	Message Name	Version	Type	Seq.
to Cboe	NewOrderJPEquitiesV1	V1	5001	Y
to Cboe	ModifyOrderJPEquitiesV1	V1	5002	Y
to Cboe	CancelOrderJPEquitiesV1	V1	5003	Y
to Cboe	PurgeOrdersJPEquitiesV1	V1	5004	Y
from Cboe	OrderAcknowledgementJPEquitiesV1	V1	5501	Y
from Cboe	OrderRejectedJPEquitiesV1	V1	5502	N
from Cboe	OrderModifiedJPEquitiesV1	V1	5503	Y
from Cboe	ModifyRejectedJPEquitiesV1	V1	5504	N
from Cboe	OrderRestatedJPEquitiesV1	V1	5505	Y
from Cboe	OrderExecutionJPEquitiesV1	V1	5506	Y
from Cboe	OrderCancelledJPEquitiesV1	V1	5507	Y
from Cboe	CancelRejectedJPEquitiesV1	V1	5508	N
from Cboe	PurgeAcknowledgementJPEquitiesV1	V1	5510	N
from Cboe	PurgeRejectedJPEquitiesV1	V1	5511	N

4.1 Participant to Cboe Japan

4.1.1 New Order

The *TradingFirm* Identifier for the order will be assigned via the port configuration.

4.1.1.1 NewOrderJPEquitiesV1

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	B0 E3 (58288)
MessageLength	2	2	Binary	97
MessageType	4	2	Binary	(5001)
MatchingUnit	6	1	Binary	Zero - from participant message
Reserved	7	1	Binary	MUST be zero from participant
SequenceNumber	8	4	Binary	*(Sequenced by Participant)*
Account	12	10	Text	Used by a trading participant to provide a broker cross reference.

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				Valid characters are ASCII 32-126.
TradingFirm	22	4	Alphanumeric	The four-character identifier of the trading firm. A zero value will be populated with the port default (see "5. Port Attributes")
ClOrdID	26	20	Text	Unique Id chosen by the participant. Characters in the ASCII range 33-126 are allowed, except for **comma, semicolon, pipe, 'at' (@)**, and **double quotes (").** If the ClOrdID matches a live order, the order will be rejected as duplicate. A leading **tilde** (~) cannot be sent on any *ClOrdID* and will result in a reject. These are reserved for internal use by Cboe Japan and could be received because of a Cboe-generated *ClOrdID*. **Note:** Although Cboe Japan only enforces uniqueness of *ClOrdID* values among currently live orders, using unique *ClOrdID* values is strongly recommended.
OrderQty	46	4	Binary	Order quantity. System limit is 999,999,999.
OrdType	50	1	Alphanumeric	2 = Limit
Price	51	8	BinaryPrice	Price *must* be an applicable minimum trading increment.
Capacity	59	1	Alphanumeric	The capacity for the order. Required on all orders. A = Agency P = Principal
Side	60	1	Alphanumeric	1 = Buy 2 = Sell 5 = Sell Short 6 = Sell Short Exempt
Symbol	61	8	Alphanumeric	Instrument symbol (QUICK symbology)
TimeInForce	69	1	Alphanumeric	0 = DAY (default) 3 = IOC (portion not filled immediately is cancelled). 4 = FOK (an IOC where the entire size must be filled immediately, otherwise the order will be cancelled back). 6 = GTD (expires at earlier of specified *ExpireTime* or end of day).
ExpireTime	70	8	DateTime	Required for GTD order (*TimeInForce* = 6), *ExpireTime* specifies the date-time (in UTC)

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				when the order expires. GTD orders will expire at the earlier of the end of the current trading day and the *ExpireTime*.
CashMargin	78	1	Alphanumeric	Indicates the type of cash or margin order. 1 = Cash 2 = Margin Open 3 = Margin Close
RiskGroupID	79	2	Binary	Used to group orders for use in order purges where multiple orders can be cancelled by specifying a list of *RiskGroupIDs* A zero value is treated as "no *RiskGroupID*", and the "Default *RiskGroupID*" port attribute value (if any) will be applied.
NoTradeFeat	81	1	Alphanumeric	Ignored if *NoTradeKey* is not set. Declares the handling of self-trading. U = No self trade protection N = Cancel Newest O = Cancel Oldest B = Cancel Both S = Cancel Smallest D = Decrement larger/Cancel Smaller d = Same as D above, but only decrement *LeavesQty*. Do not restate *OrderQty*. NULL (0x00) = Apply the port default. If the older and newer orders in a self-trade have conflicting instructions, the instruction of the newer order will be honored, except that if the inbound order specifies Decrement and the resting order does not, and the resting order is larger, then both orders will be cancelled. This exception is to protect the order entry software for the resting order from receiving an unexpected restatement message.
NoTradeKey	82	10	Text	Interactions of orders from the same participant with the same *NoTradeKey* value will be managed by the newer order's *NoTradeFeat* instruction.
PostingInstruction	92	1	Alpha	NUL (0x00) = allow taking P = Post Only (never take liquidity)
OrderClassification	93	1	Alphanumeric	Specify the order classification. Any alphanumeric character (or NULL) is accepted.

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				Suggested values: NULL (0x00) = not specified 1 = Non Low Latency trading (Auto) 2 = Non Low Latency trading (Manual) 3 = Market Making Strategy 4 = Arbitrage Strategy 5 = Directional Strategy 6 = Other Strategy
MarginTransactionType	94	1	Alphanumeric	Specifies if a margin order is a Negotiable or a Standardized. 1 = Negotiable 2 = Standardized NULL (0x00) = None Required if *CashMargin is 2* (Margin Open) or 3 (Margin Close), otherwise ignored.
CompanyID	95	4	Alphanumeric	Up to 4 alphanumeric characters to identify the order entry firm. This field is not validated.

4.1.2 Modify Order

Request to modify an order. Only *Price* and *OrderQty* may be adjusted. Modifies will result in a loss of time priority unless the modification involves only a decrease in *OrderQty*.

Changes in *OrderQty* results in an adjustment of the current order's *OrderQty*. The new *OrderQty* does not directly replace the current order's *LeavesQty*. Rather, a delta is computed from the current *OrderQty* and the replacement *OrderQty*. This delta is then applied to the current *LeavesQty*. If the resulting *LeavesQty* is less than or equal to zero, the order is cancelled. This results in safer behavior when the modification request overlaps partial fills for the current order, leaving the Participant in total control of the share exposure of the order.

A maximum of 1,679,615 `Modify Order` requests may be made to a single order each trading day. Once the 1,679,615th modification is made, then the next user-generated message on the order should be a `Cancel Order` request.

4.1.2.1 ModifyOrderJPEquitiesV1

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	B0 E3 (58288)
MessageLength	2	2	Binary	67
MessageType	4	2	Binary	(5002)
MatchingUnit	6	1	Binary	Zero - from participant message
Reserved	7	1	Binary	MUST be zero from participant

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SequenceNumber	8	4	Binary	*(Sequenced by Participant)*
TradingFirm	12	4	Alphanumeric	The four-character identifier of the trading firm.
ClOrdID	16	20	Text	Client Order Identifier for the modified order. Unique Id chosen by the participant. Characters in the ASCII range 33-126 are allowed, except for **comma, semicolon, pipe, 'at' (@)**, and **double quotes (").** If the ClOrdID matches a live order, the order will be rejected as duplicate. A leading **tilde** (~) cannot be sent on any *ClOrdID* and will result in a reject. These are reserved for internal use by Cboe Japan and could be received because of a Cboe-generated *ClOrdID*. **Note:** Although Cboe Japan only enforces uniqueness of *ClOrdID* values among currently live orders, using unique *ClOrdID* values is strongly recommended.
OrderQty	36	4	Binary	Order quantity. System limit is 999,999,999.
OrigClOrdID	40	20	Text	The *ClOrdID* of the original order. In the case of multiple changes to a single order, this will be the *ClOrdID* of the **most recent** accepted change.
Price	60	8	BinaryPrice	For the original order's *OrdType*, Price *must* be an applicable minimum trading increment.
CancelOrigOnReject	68	1	Alpha	Y = Cancel original order if modification fails. N = Leave original order alone. NULL (0x00) = Apply port default

4.1.3 Cancel Order

Request to cancel a single order.

4.1.3.1 CancelOrderJPEquitiesV1

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	B0 E3 (58288)
MessageLength	2	2	Binary	34
MessageType	4	2	Binary	(5003)

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MatchingUnit	6	1	Binary	Zero - from participant message
Reserved	7	1	Binary	MUST be zero from participant
SequenceNumber	8	4	Binary	*(Sequenced by Participant)*
TradingFirm	12	4	Alphanumeric	The four-character identifier of the trading firm. A zero value will be populated with the port default (see "5. Port Attributes").
OrigClOrdID	16	20	Text	The *ClOrdID* of the original order.

4.1.4 Purge Orders

Request to cancel a group of orders across all of the participant's sessions. The `Purge Orders` message will only be accepted dedicated purge ports (see "1.5.4.3 - BOE Purge Ports").

Participants may choose to implement one or more of the following filters:

1. *Trading Firm* Filter – optionally cancel based on the Trading Firm identifier. Set using first character of *MassCancelInst* and sending *TradingFirm*.
2. *Symbol* Filter – optionally cancel based on symbol. Set by sending a valid symbol in the *Symbol* fields. Cannot be combined with *RiskGroupID* filter.
3. *RiskGroupID* Filter – optionally cancel based on *RiskGroupID.* A maximum of 10 *RiskGroupIDs* may be included on a single `Purge Request` message. Set by populating *RiskGroupIDCnt to* a non-zero value. Cannot be combined with *Symbol* filter.

Participants may use the second character of *MassCancelInst* to set the acknowledgement style. If a single `Purge Acknowledgement` is selected, then *MassCancelID* must be sent.

The system limits the rate at which identical `Purge Request` messages can be submitted to the system. Requests are restricted to twenty (20) messages per second per port.

An identical purge message is defined as a message having all the same *RiskGroupID*, *Symbol*, and *TradingFirm* field values as a previously received message.

4.1.4.1 PurgeOrdersJPEquitiesV1

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	B0 E3 (58288)
MessageLength	2	2	Binary	59 + (*RiskGroupIDCnt* * 2)
MessageType	4	2	Binary	(5004)
MatchingUnit	6	1	Binary	Zero - from participant message
Reserved	7	1	Binary	MUST be zero from participant
SequenceNumber	8	4	Binary	*(Sequenced by Participant)*

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TradingFirm	12	4	Alphanumeric	The four-character identifier of the trading firm.
MassCancelInst	16	16	Alphanumeric	At least one character must be provided (Firm Filter). Contiguous characters must be specified up to total length. Truncated (unspecified) characters will default to values indicated below. ***1st Character : Firm Filter*** A = No filtering by firm is performed. F = All orders that were sent under the firm specified in *TradingFirm* will be cancelled. ***2nd Character : Acknowledgement Style*** M = (default) Individual Execution Reports are sent for each cancelled order. S = Single Purge Acknowledgement sent once all cancels have been processed. Single Purge Acknowledgement will contain *MassCancelID* and *CancelledOrderCount*. *MassCancelID* must be specified, or the Purge Request will be rejected. B = Both individual Execution Reports and single Purge Acknowledgement will be sent. Also requires *MassCancelID* to be specified or the Purge Request will be rejected.
MassCancelID	32	20	Text	User-defined identifier of the mass cancel or purge request.
Symbol	52	8	Alphanumeric	Instrument symbol (QUICK symbology)
RiskGroupIDCnt	60	1	Binary	Number of repeating *RiskGroupID* included in this message. Value 0 to 10.
➔ *RiskGroupID*	61	2	Binary	Used to group orders for use in order purges where multiple orders can be cancelled by specifying a list of *RiskGroupIDs*

4.2 Cboe Japan to Participant

4.2.1 Order Acknowledgement

`Order Acknowledgement` messages are sent in response to a `New Order` message. The message corresponds to a FIX Execution Report with *ExecType* (150) = 0 (New). `Order Acknowledgement` messages do not indicate how or if the order was executed or cancelled. Note that the `Order Acknowledgement` of an IOC order (*TimeInForce* = 3) will be immediately followed by zero or more `Order Execution` (fill) messages and at most one `Order Cancelled` message (for any remaining size).

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4.2.1.1 OrderAcknowledgementJPEquitiesV1

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	B0 E3 (58288)
MessageLength	2	2	Binary	114
MessageType	4	2	Binary	(5501)
MatchingUnit	6	1	Binary	
Reserved	7	1	Binary	Value sent to participant not specified
SequenceNumber	8	4	Binary	*(Sequenced per MatchingUnit by Cboe Japan)*
TransactionTime	12	8	DateTime	The time the event occurred in the Matching Engine (not the time the message was sent).
Account	20	10	Text	(Copied from order)
TradingFirm	30	4	Alphanumeric	(Copied from order / port default)
ClOrdID	34	20	Text	(Copied from order)
OrderID	54	8	Binary	Order identifier supplied by Cboe Japan. This identifier corresponds to the identifiers used in Cboe Japan market data products.
OrderQty	62	4	Binary	(Copied from order)
OrdType	66	1	Alphanumeric	(Copied from order)
Price	67	8	BinaryPrice	(Copied from order)
Capacity	75	1	Alphanumeric	(Copied from order)
Side	76	1	Alphanumeric	(Copied from order)
Symbol	77	8	Alphanumeric	(Copied from order)
TimeInForce	85	1	Alphanumeric	(Copied from order)
ExpireTime	86	8	DateTime	(Copied from order)
CashMargin	94	1	Alphanumeric	(Copied from order)
RiskGroupID	95	2	Binary	(Copied from order / port default)
NoTradeFeat	97	1	Alphanumeric	(Copied from order / port default)
NoTradeKey	98	10	Text	(Copied from order)
PostingInstruction	108	1	Alpha	(Copied from order)
OrderClassification	109	1	Alphanumeric	(Copied from order)
MarginTransactionType	110	1	Alphanumeric	(Copied from order)
BaseLiquidityIndicator	111	1	Alphanumeric	Indicates whether the trade added or removed liquidity. A = Added Liquidity

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				R = Removed Liquidity
CompanyID	112	4	Alphanumeric	Up to 4 alphanumeric characters to identify the order entry firm. This field is not validated.

(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)

4.2.2 Order Rejected

`Order Rejected` messages are sent in response to a `New Order` which must be rejected. This message corresponds to a FIX Execution Report with *ExecType* (150) = 8 (Rejected). `Order Rejected` messages are unsequenced.

4.2.2.1 OrderRejectedJPEquitiesV1

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	B0 E3 (58288)
MessageLength	2	2	Binary	103
MessageType	4	2	Binary	(5502)
MatchingUnit	6	1	Binary	
Reserved	7	1	Binary	Value sent to participant not specified
SequenceNumber	8	4	Binary	*(Unsequenced: always 0)*
TransactionTime	12	8	DateTime	The time the event occurred in the Matching Engine (not the time the message was sent).
TradingFirm	20	4	Alphanumeric	The four-character identifier of the trading firm.
ClOrdID	24	20	Text	Client order identifier of the rejected order.
OrderRejectReason	44	1	Text	See Section '4.5 - Order Reason Codes'.
Text	45	60	Text	Human readable text with more information.

(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)

4.2.3 Order Modified

`Order Modified` messages are sent in response to a `Modify Order` to indicate that the order has been successfully modified.

In some cases, the last message to be received on an order's lifecycle will be an `Order Modified` message. The way to know the order is no longer live is to inspect *LeavesQty*. An example of this would be modification of an order while an execution is being generated, resulting in the order being reduced to zero outstanding quantity.

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4.2.3.1 OrderModifiedJPEquitiesV1

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	B0 E3 (58288)
MessageLength	2	2	Binary	91
MessageType	4	2	Binary	(5503)
MatchingUnit	6	1	Binary	
Reserved	7	1	Binary	Value sent to participant not specified
SequenceNumber	8	4	Binary	*(Sequenced per MatchingUnit by Cboe Japan)*
TransactionTime	12	8	DateTime	The time the event occurred in the Matching Engine (not the time the message was sent).
TradingFirm	20	4	Alphanumeric	The four-character identifier of the trading firm.
ClOrdID	24	20	Text	The *ClOrdID* of the modified order.
OrderQty	44	4	Binary	Order quantity. System limit is 999,999,999.
OrigClOrdID	48	20	Text	The *ClOrdID* of the original order.
OrderID	68	8	Binary	Order identifier supplied by Cboe Japan. This identifier corresponds to the identifiers used in Cboe Japan market data products.
Price	76	8	BinaryPrice	The limit or peg price of the order after modification.
LeavesQty	84	4	Binary	Quantity still open for further execution. If zero, the order is complete.
BaseLiquidityIndicator	88	1	Alphanumeric	Indicates whether the trade added or removed liquidity. A = Added Liquidity R = Removed Liquidity
CompanyID	89	4	Alphanumeric	Up to 4 alphanumeric characters to identify the order entry firm. This field is not validated.

(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)

4.2.4 Modify Rejected

`Modify Rejected` messages are sent in response to a `Modify Order` for an order which cannot be modified. `Modify Rejected` messages are unsequenced.

This message corresponds to a FIX Execution Report with *MsgType* (35) = 9 (Order Cancel Reject) and *CxlRejResponseTo* (434) = 2 (Order Cancel/Replace Request).

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4.2.4.1 ModifyRejectedJPEquitiesV1

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	B0 E3 (58288)
MessageLength	2	2	Binary	123
MessageType	4	2	Binary	(5504)
MatchingUnit	6	1	Binary	
Reserved	7	1	Binary	Value sent to participant not specified
SequenceNumber	8	4	Binary	*(Unsequenced: always 0)*
TransactionTime	12	8	DateTime	The time the event occurred in the Matching Engine (not the time the message was sent).
TradingFirm	20	4	Alphanumeric	The four-character identifier of the trading firm.
ClOrdID	24	20	Text	ClOrdID of the Modify Order.
OrigClOrdID	44	20	Text	The *ClOrdID* of the original order.
ModifyRejectReason	64	1	Text	See Section '4.5 - Order Reason Codes'.
Text	65	60	Text	Human readable text with more information.

(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)

4.2.5 Order Restated

If an order is cancelled or reduced due to Self-Trade Protection instructions (see *NoTradeFeat*), the order will be restated with reductions to *OrderQty* and/or *LeavesQty*. Note that *LeavesQty* may be reduced to 0, indicating that the order has effectively been cancelled for Self-Trade Protection.

4.2.5.1 OrderRestatedJPEquitiesV1

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	B0 E3 (58288)
MessageLength	2	2	Binary	88
MessageType	4	2	Binary	(5505)
MatchingUnit	6	1	Binary	
Reserved	7	1	Binary	Value sent to participant not specified
SequenceNumber	8	4	Binary	*(Sequenced per MatchingUnit by Cboe Japan)*
TransactionTime	12	8	DateTime	The time the event occurred in the Matching Engine (not the time the message was sent).
TradingFirm	20	4	Alphanumeric	The four-character identifier of the trading firm.

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ClOrdID	24	20	Text	The Client Order ID of the order being restated.
OrderID	44	8	Binary	Order identifier supplied by Cboe Japan. This identifier corresponds to the identifiers used in Cboe Japan market data products.
OrderQty	52	4	Binary	Order quantity. This may be less than the original OrderQty on the order.
Price	56	8	BinaryPrice	The Limit or Cap Price of the order.
LeavesQty	64	4	Binary	Quantity still open for further execution. If zero, the order is complete.
SecondaryOrderID	68	8	Binary	Cboe *OrderID* of contra side of prevented match.
LastPx	76	8	BinaryPrice	The price of the execution avoided by Self-Trade Protection. Equivalent to *PreventedTradePx* (7903) in FIX.
LastShares	84	4	Binary	If *RestatementReason*=W, this is the size of the execution avoided by Self-Trade Protection. Equivalent to *PreventedTradeLiquidityIndicator* (7905) in FIX. Otherwise, zero.
BaseLiquidityIndicator	88	1	Alphanumeric	For the trade avoided by self-trade, indicates whether the trade would have added or removed order liquidity. Equivalent to *PreventedTradePx* (7903) in FIX. `A` = Added Liquidity `R` = Removed Liquidity `NULL (0x00)` = n/a
RestatementReason	89	1	Alphanumeric	The reason for this Order Restated message. `W` = Would have self-traded

4.2.6 Order Execution

An `Order Execution` is sent for each fill on an order.

Rather than returning a monetary value indicating the rebate or charge for an execution, the *FeeCode* is an indication of a fee classification corresponding to an item on the venue's fee schedule.

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4.2.6.1 OrderExecutionJPEquitiesV1

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	B0 E3 (58288)
MessageLength	2	2	Binary	77
MessageType	4	2	Binary	(5506)
MatchingUnit	6	1	Binary	
Reserved	7	1	Binary	Value sent to participant not specified
SequenceNumber	8	4	Binary	*(Sequenced per MatchingUnit by Cboe Japan)*
TransactionTime	12	8	DateTime	The time the event occurred in the Matching Engine (not the time the message was sent).
TradingFirm	20	4	Alphanumeric	The four-character identifier of the trading firm.
ClOrdID	24	20	Text	The Client Order ID of the traded order.
ExecID	44	8	Binary	Execution Id. Unique across all matching units on a given day. Note: ExecIDs will be represented on ODROP and FIXDROP ports as base 36 ASCII.
LastCapacity	52	1	Alphanumeric	Capacity of the order. 1 = Agent 2 = Cross as Agent 3 = Cross as Principal 4 = Principal Values of 2 and 3 indicate the client has executed against themselves.
LastPx	53	8	BinaryPrice	Price of this fill.
LastShares	61	4	Binary	Executed quantity.
Side	65	1	Alphanumeric	1 = Buy 2 = Sell 5 = Sell Short 6 = Sell Short Exempt
Symbol	66	8	Alphanumeric	Instrument symbol (QUICK symbology)
LeavesQty	74	4	Binary	Quantity still open for further execution. If zero, the order is complete.
BaseLiquidityIndicator	78	1	Alphanumeric	Indicates whether the trade added or removed liquidity. A = Added Liquidity R = Removed Liquidity

(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)

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4.2.7 Order Cancelled

4.2.7.1 OrderCancelledJPEquitiesV1

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	B0 E3 (58288)
MessageLength	2	2	Binary	43
MessageType	4	2	Binary	(5507)
MatchingUnit	6	1	Binary	
Reserved	7	1	Binary	Value sent to participant not specified
SequenceNumber	8	4	Binary	*(Sequenced per MatchingUnit by Cboe Japan)*
TransactionTime	12	8	DateTime	The time the event occurred in the Matching Engine (not the time the message was sent).
TradingFirm	20	4	Alphanumeric	The four-character identifier of the trading firm.
ClOrdID	24	20	Text	Client Order ID of the cancelled order.
CancelReason	44	1	Text	See Section '4.5 - Order Reason Codes'.

(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)

4.2.8 Cancel Rejected

A `Cancel Rejected` message is sent in response to a `Cancel Order` message to indicate that the cancellation cannot occur. `Cancel Rejected` messages are unsequenced.

4.2.8.1 CancelRejectedJPEquitiesV1

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	B0 E3 (58288)
MessageLength	2	2	Binary	103
MessageType	4	2	Binary	(5508)
MatchingUnit	6	1	Binary	
Reserved	7	1	Binary	Value sent to participant not specified
SequenceNumber	8	4	Binary	*(Unsequenced: always 0)*
TransactionTime	12	8	DateTime	The time the event occurred in the Matching Engine (not the time the message was sent).
TradingFirm	20	4	Alphanumeric	The four-character identifier of the trading firm.
ClOrdID	24	20	Text	The Client Order ID of the `Cancel Order` message.

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CancelRejectReason	44	1	Text	See Section '4.5 - Order Reason Codes'.
Text	45	60	Text	Human readable text with more information.

(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)

4.2.9 Purge Acknowledgement

A `Purge Acknowledgment` is an unsequenced message sent when a `Purge Orders` message requesting an order purge has completed cancelling all individual orders.

4.2.9.1 PurgeAcknowledgementJPEquitiesV1

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	B0 E3 (58288)
MessageLength	2	2	Binary	42
MessageType	4	2	Binary	(5510)
MatchingUnit	6	1	Binary	
Reserved	7	1	Binary	Value sent to participant not specified
SequenceNumber	8	4	Binary	*(Unsequenced: always 0)*
TransactionTime	12	8	DateTime	The time the event occurred in the Matching Engine (not the time the message was sent).
MassCancelID	20	20	Text	User-defined identifier of the mass cancel or purge request.
CancelledOrderCount	40	4	Binary	Number of orders cancelled.

(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)

4.2.10 Purge Rejected

A `Purge Rejected` message is sent in response to a `Purge Orders` message to indicate that the mass cancellation cannot occur. `Purge Rejected` messages are unsequenced.

4.2.10.1 PurgeRejectedJPEquitiesV1

Field Name	Offset	Length	Data Type	Description
StartOfMessage	0	2	Binary	B0 E3 (58288)
MessageLength	2	2	Binary	99
MessageType	4	2	Binary	(5511)
MatchingUnit	6	1	Binary	
Reserved	7	1	Binary	Value sent to participant not specified
SequenceNumber	8	4	Binary	*(Unsequenced: always 0)*

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TransactionTime	12	8	DateTime	The time the event occurred in the Matching Engine (not the time the message was sent).
MassCancelID	20	20	Text	User-defined identifier of the mass cancel or purge request.
PurgeRejectReason	40	1	Text	Reason for the purge rejection. See Section '4.5 - Order Reason Codes'.
Text	41	60	Text	Human readable text with more information.

(New fields may be introduced at the end of this message. Consequently, Participants must treat any additional bytes present as undefined values.)

4.3 Application Example Messages

4.3.1 NewOrderJPEquitiesV1

Field	**Hex Bytes**	**Value**
StartOfMessage	b0 e3	BOE3 (58288, 0xe3b0)
MessageLength	61 00	97
MessageType	89 13	NewOrderJPEquitiesV1 (5001, 0x1389)
MatchingUnit	00	0
Reserved	00	0
SequenceNumber	47 94 03 00	234567
Account10	61 62 63 64 2d 31 20 39 5a 00	abcd-1 9Z
TradingFirm	31 32 33 34	1234
ClOrdID	41 42 43 44 2d 39 39 39 30 00 00 00 00 00 00 00 00 00 00 00	ABCD-9990
OrderQty	dc 05 00 00	1500
OrdType	32	Limit (50, 0x32, '2')
Price	58 86 03 00 00 00 00 00	23.10
Capacity	41	Agency (65, 0x41, 'A')
Side	32	Sell (50, 0x32, '2')
Symbol	34 33 32 31 00 00 00 00	4321
TimeInForce	36	GTD (54, 0x36, '6')
ExpireTime	00 10 83 c6 c6 86 32 17	2022-12-20 14:30:00.000000
CashMargin	32	MarginOpen (50, 0x32, '2')
RiskGroupID	12 00	18
NoTradeFeat	4e	CancelNewest (78, 0x4e, 'N')
NoTradeKey	41 42 43 31 32 33 00 00 00 00	ABC123
PostingInstruction	00	
OrderClassification	34	4
MarginTransactionType	32	Standardized (50, 0x32, '2')
CompanyID	63 69 64 39	cid9

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4.3.2 ModifyOrderJPEquitiesV1

Field	Hex Bytes	Value
StartOfMessage	b0 e3	BOE3 (58288, 0xe3b0)
MessageLength	43 00	67
MessageType	8a 13	ModifyOrderJPEquitiesV1 (5002, 0x138a)
MatchingUnit	00	0
Reserved	00	0
SequenceNumber	48 94 03 00	234568
TradingFirm	31 32 33 34	1234
ClOrdID	41 42 43 44 2d 39 39 39 31 00 00 00 00 00 00 00 00 00 00 00	ABCD-9991
OrderQty	f0 0a 00 00	2800
OrigClOrdID	41 42 43 44 2d 39 39 39 30 00 00 00 00 00 00 00 00 00 00 00	ABCD-9990
Price	f0 c0 03 00 00 00 00 00	24.60
CancelOrigOnReject	59	Yes (89, 0x59, 'Y')

4.3.3 CancelOrderJPEquitiesV1

Field	Hex Bytes	Value
StartOfMessage	b0 e3	BOE3 (58288, 0xe3b0)
MessageLength	22 00	34
MessageType	8b 13	CancelOrderJPEquitiesV1 (5003, 0x138b)
MatchingUnit	00	0
Reserved	00	0
SequenceNumber	49 94 03 00	234569
TradingFirm	31 32 33 34	1234
OrigClOrdID	41 42 43 44 2d 39 39 39 30 00 00 00 00 00 00 00 00 00 00 00	ABCD-9990

4.3.4 PurgeOrdersJPEquitiesV1

Field	Hex Bytes	Value
StartOfMessage	b0 e3	BOE3 (58288, 0xe3b0)
MessageLength	3b 00	59
MessageType	8c 13	PurgeOrdersJPEquitiesV1 (5004, 0x138c)
MatchingUnit	00	0
Reserved	00	0
SequenceNumber	4a 94 03 00	234570
TradingFirm	31 32 33 34	1234
MassCancelInst	46 42 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	FB
MassCancelID	6d 61 73 73 63 78 6c 2d 35 35 34 34 00 00 00 00 00 00 00 00	masscxl-5544

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Symbol	32 33 34 35 00 00 00 00	2345
CustomGroupIDCnt	00	0

4.3.5 PurgeOrdersJPEquitiesV1

Field	**Hex Bytes**	**Value**
StartOfMessage	b0 e3	BOE3 (58288, 0xe3b0)
MessageLength	3f 00	63
MessageType	8c 13	PurgeOrdersJPEquitiesV1 (5004, 0x138c)
MatchingUnit	00	0
Reserved	00	0
SequenceNumber	4b 94 03 00	234571
TradingFirm	31 32 33 34	1234
MassCancelInst	46 42 00 00 00 00 00 00 00 00 00 00 00 00 00 00	FB
MassCancelID	6d 61 73 73 63 78 6c 2d 35 35 34 41 00 00 00 00 00 00 00 00	masscxl-554A
Symbol	00 00 00 00 00 00 00 00	
RiskGroupIDCnt	02	2
RiskGroupId[1/2]	12 00	18
RiskGroupId[2/2]	1e 01	286

4.3.6 OrderAcknowledgementJPEquitiesV1

Field	**Hex Bytes**	**Value**
StartOfMessage	b0 e3	BOE3 (58288, 0xe3b0)
MessageLength	72 00	114
MessageType	7d 15	OrderAcknowledgementJPEquitiesV1 (5501, 0x157d)
MatchingUnit	01	1
Reserved	00	0
SequenceNumber	f1 fb 09 00	654321
TransactionTime	28 a8 4a 1c 88 83 32 17	2022-12-20 13:30:32.321321
Account10	61 62 63 64 2d 31 20 39 5a 00	abcd-1 9Z
TradingFirm	31 32 33 34	1234
ClOrdID	41 42 43 44 2d 39 39 39 30 00 00 00 00 00 00 00 00 00 00 00	ABCD-9990
OrderID	3d 07 7a 13 ee 0e 34 02	17FANZAA3D4T
OrderQty	dc 05 00 00	1500
OrdType	32	Limit (50, 0x32, '2')
Price	58 86 03 00 00 00 00 00	23.10
Capacity	41	Agency (65, 0x41, 'A')
Side	32	Sell (50, 0x32, '2')
Symbol	34 33 32 31 00 00 00 00	4321
TimeInForce	36	GTD (54, 0x36, '6')
ExpireTime	00 10 83 c6 c6 86 32 17	2022-12-20 14:30:00.000000
CashMargin	32	MarginOpen (50, 0x32, '2')

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RiskGroupID	12 00	18
NoTradeFeat	4e	CancelNewest (78, 0x4e, 'N')
NoTradeKey	41 42 43 31 32 33 00 00 00 00	ABC123
PostingInstruction	00	
OrderClassification	34	4
MarginTransactionType	32	Standardized (50, 0x32, '2')
BaseLiquidityIndicator	41	Added (65, 0x41, 'A')
CompanyID	63 69 64 39	cid9

4.3.7 OrderRejectedJPEquitiesV1

Field	**Hex Bytes**	**Value**
StartOfMessage	b0 e3	BOE3 (58288, 0xe3b0)
MessageLength	67 00	103
MessageType	7e 15	OrderRejectedJPEquitiesV1 (5502, 0x157e)
MatchingUnit	00	0
Reserved	00	0
SequenceNumber	00 00 00 00	0
TransactionTime	28 72 e5 57 88 83 32 17	2022-12-20 13:30:33.321321
TradingFirm	31 32 33 34	1234
ClOrdID	41 42 43 44 2d 39 39 39 30 00 00 00 00 00 00 00 00 00 00 00	ABCD-9990
OrderRejectReason	41	A
Text	54 45 53 54 20 54 45 53 54 20 54 45 53 54 00 00 00 00 00 00 00 00 00 00 00

	00 00 00 00 00 00 00 00 00 00 00 00	
OrderID	3d 07 7a 13 ee 0e 34 02	17FANZAA3D4T
Price	f0 c0 03 00 00 00 00 00	24.60
LeavesQty	f4 01 00 00	500
BaseLiquidityIndicator	52	Removed (82, 0x52, 'R')
CompanyID	63 69 64 39	cid9

4.3.9 ModifyRejectedJPEquitiesV1

Field	**Hex Bytes**	**Value**
StartOfMessage	b0 e3	BOE3 (58288, 0xe3b0)
MessageLength	7b 00	123
MessageType	80 15	ModifyRejectedJPEquitiesV1 (5504, 0x1580)
MatchingUnit	00	0
Reserved	00	0
SequenceNumber	00 00 00 00	0
TransactionTime	28 06 1b cf 88 83 32 17	2022-12-20 13:30:35.321321
TradingFirm	31 32 33 34	1234
ClOrdID	41 42 43 44 2d 39 39 39 31 00 00 00 00 00 00 00 00 00 00 00	ABCD-9991
OrigClOrdID	41 42 43 44 2d 39 39 39 30 00 00 00 00 00 00 00 00 00 00 00	ABCD-9990
ModifyRejectReason	41	A
Text	54 45 53 54 20 54 45 53 54 20 54 45 53 54 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	TEST TEST TEST

4.3.10 OrderRestatedJPEquitiesV1

Field	**Hex Bytes**	**Value**
StartOfMessage	b0 e3	BOE3 (58288, 0xe3b0)
MessageLength	58 00	88
MessageType	81 15	OrderRestatedJPEquitiesV1 (5505, 0x1581)
MatchingUnit	01	1
Reserved	00	0
SequenceNumber	f3 fb 09 00	654323
TransactionTime	28 dc 2c 4d 8c 83 32 17	2022-12-20 13:30:50.321321
TradingFirm	31 32 33 34	1234
ClOrdID	41 42 43 44 2d 39 39 39 30 00 00 00 00 00 00 00 00 00 00 00	ABCD-9990

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OrderID	3d 07 7a 13 ee 0e 34 02	17FANZAA3D4T
OrderQty	dc 05 00 00	1500
Price	58 86 03 00 00 00 00 00	23.10
LeavesQty	f4 01 00 00	500
SecondaryOrderID	e6 9c 83 13 ee 0e 34 02	17FANZAAGTT2
LastPx	58 86 03 00 00 00 00 00	23.10
LastShares	2c 01 00 00	300
BaseLiquidityIndicator	41	Added (65, 0x41, 'A')
RestatementReason	57	Wash (87, 0x57, 'W')

4.3.11 OrderExecutionJPEquitiesV1

Field	**Hex Bytes**	**Value**
StartOfMessage	b0 e3	BOE3 (58288, 0xe3b0)
MessageLength	4d 00	77
MessageType	82 15	OrderExecutionJPEquitiesV1 (5506, 0x1582)
MatchingUnit	01	1
Reserved	00	0
SequenceNumber	f4 fb 09 00	654324
TransactionTime	28 a6 24 5b b9 79 c2 16	2021-12-20 13:30:51.321321
TradingFirm	31 32 33 34	1234
ClOrdID	41 42 43 44 2d 39 39 39 30 00 00 00 00 00 00 00 00 00 00 00	ABCD-9990
ExecID	79 57 de 3e 12 00 00 00	0100004MX
LastCapacity	31	Agent (49, 0x31, '1')
LastPx	58 86 03 00 00 00 00 00	23.10
LastShares	f4 01 00 00	500
Side	32	Sell (50, 0x32, '2')
Symbol	34 33 32 31 00 00 00 00	4321
LeavesQty	00 00 00 00	0
BaseLiquidityIndicator	41	Added (65, 0x41, 'A')

4.3.12 OrderCancelledJPEquitiesV1

Field	**Hex Bytes**	**Value**
StartOfMessage	b0 e3	BOE3 (58288, 0xe3b0)
MessageLength	2b 00	43
MessageType	83 15	OrderCancelledJPEquitiesV1 (5507, 0x1583)
MatchingUnit	01	1
Reserved	00	0
SequenceNumber	f5 fb 09 00	654325
TransactionTime	28 70 62 c4 8c 83 32 17	2022-12-20 13:30:52.321321
TradingFirm	31 32 33 34	1234
ClOrdID	41 42 43 44 2d 39 39 39 30 00 00 00 00 00 00 00 00 00 00 00	ABCD-9990
CancelReason	41	A

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4.3.13 CancelRejectedJPEquitiesV1

Field	Hex Bytes	Value
StartOfMessage	b0 e3	BOE3 (58288, 0xe3b0)
MessageLength	67 00	103
MessageType	84 15	CancelRejectedJPEquitiesV1 (5508, 0x1584)
MatchingUnit	00	0
Reserved	00	0
SequenceNumber	00 00 00 00	0
TransactionTime	28 d0 b5 0a 89 83 32 17	2022-12-20 13:30:36.321321
TradingFirm	31 32 33 34	1234
ClOrdID	41 42 43 44 2d 39 39 39 31 00 00 00 00 00 00 00 00 00 00 00	ABCD-9991
CancelRejectReason	41	A
Text	54 45 53 54 20 54 45 53 54 20 54 45 53 54 00 00 00 00 00 00 00 00 00 00 00

MassCancelID	6d 61 73 73 63 78 6c 2d 35 35 34 41 00 00 00 00 00 00 00 00	masscxl-554A
PurgeRejectReason	41	A
Text	54 45 53 54 20 54 45 53 54 20 54 45 53 54 00 00 00 00 00 00 00 00

CashMargin	544	1	Alphanumeric	Indicates the type of cash or margin order. 1 = Cash 2 = Margin Open 3 = Margin Close
ClOrdID	11	20	Text	Unique Id chosen by the participant. Characters in the ASCII range 33-126 are allowed, except for **comma, semicolon, pipe, 'at' (@)**, and **double quotes (").** If the ClOrdID matches a live order, the order will be rejected as duplicate. A leading **tilde** (~) cannot be sent on any *ClOrdID* and will result in a reject. These are reserved for internal use by Cboe Japan and could be received because of a Cboe-generated *ClOrdID*. **Note:** Although Cboe Japan only enforces uniqueness of *ClOrdID* values among currently live orders, using unique *ClOrdID* values is strongly recommended.
CompanyID	-	4	Alphanumeric	Up to 4 alphanumeric characters to identify the order entry firm. This field is not validated.
ExecID	17	8	Binary	Execution Id. Unique across all matching units on a given day. Note: ExecIDs will be represented on ODROP and FIXDROP ports as base 36 ASCII. *
ExpireTime	424	8	DateTime	Required for GTD order (*TimeInForce* = 6), *ExpireTime* specifies the date-

* Example conversion:

Decimal	Base 36
28294005440239	A1234B567
76335905726621	R248BC23H
728557228187	09AP05V2Z

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				time (in UTC) when the order expires. GTD orders will expire at the earlier of the end of the current trading day and the *ExpireTime*.
FeeCode	9882	2	Alphanumeric	Indicates fee associated with an execution. Fee codes are published in the pricing schedule. New fee codes may be sent with little notice. Participants are encouraged to code their systems to accept unknown fee codes.
LastCapacity	29	1	Alphanumeric	Capacity of the order. 1 = Agent 2 = Cross as Agent 3 = Cross as Principal 4 = Principal Values of 2 and 3 indicate the client has executed against themselves.
LastPx	31	8	BinaryPrice	Price of this fill.
LastShares	32	4	Binary	Executed quantity.
LeavesQty	151	4	Binary	Quantity still open for further execution. If zero, the order is complete.
MarginTransactionType	8214	1	Alphanumeric	Specifies if a margin order is a Negotiable or a Standardized. 1 = Negotiable 2 = Standardized NULL (0x00) = None Required if *CashMargin* is 2 (Margin Open) or 3 (Margin Close), otherwise ignored.
MassCancelID	7695	20	Alphanumeric	User-defined identifier of the mass cancel or purge request.
MassCancelInst	7700	16	Alphanumeric	At least one character must be provided (Firm Filter). Contiguous characters must be specified up to total length. Truncated (unspecified) characters will default to values indicated below. ***1st Character : Firm Filter*** A = No filtering by firm is performed.

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				F = All orders that were sent under the firm specified in *TradingFirm* will be cancelled. ***2nd Character : Acknowledgement Style*** M = (default) Individual Execution Reports are sent for each cancelled order. S = Single Purge Acknowledgement sent once all cancels have been processed. Single Purge Acknowledgement will contain *MassCancelID* and *CancelledOrderCount*. *MassCancelID* must be specified, or the Purge Request will be rejected. B = Both individual Execution Reports and single Purge Acknowledgement will be sent. Also requires *MassCancelID* to be specified or the Purge Request will be rejected.
ModifyRejectReason	103	1	Text	See Section '4.5 - Order Reason Codes'.
NoTradeFeat	7713	1	Alphanumeric	Ignored if *NoTradeKey* is not set. Declares the handling of self-trading. U = No self trade protection N = Cancel Newest O = Cancel Oldest B = Cancel Both S = Cancel Smallest D = Decrement larger/Cancel Smaller d = Same as D above, but only decrement *LeavesQty*. Do not restate *OrderQty*. NULL (0x00) = Apply the port default. If the older and newer orders in a self-trade have conflicting instructions, the instruction of the newer order will be honored, except that if the inbound order specifies Decrement and the resting order

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				does not, and the resting order is larger, then both orders will be cancelled. This exception is to protect the order entry software for the resting order from receiving an unexpected restatement message.
NoTradeKey	7714	10	Text	Interactions of orders from the same participant with the same *NoTradeKey* value will be managed by the newer order's *NoTradeFeat* instruction.
OrderClassification	8060	1	Alphanumeric	Specify the order classification. Any alphanumeric character (or NULL) is accepted. Suggested values: `NULL (0x00)` = not specified `1` = Non Low Latency trading (Auto) `2` = Non Low Latency trading (Manual) `3` = Market Making Strategy `4` = Arbitrage Strategy `5` = Directional Strategy `6` = Other Strategy
OrderID	37	8	Binary	Order identifier supplied by Cboe Japan. This identifier corresponds to the identifiers used in Cboe Japan market data products.
OrderQty	31	4	Binary	Order quantity. System limit is 999,999,999.
OrderRejectReason	103	1	Text	See Section '4.5 - Order Reason Codes'.
OrdType	40	1	Alphanumeric	`2` = Limit
OrigClOrdID	41	20	Text	The *ClOrdID* of the original order.
PostingInstruction	9303	1	Alpha	NUL (0x00) = allow taking P = Post Only (never take liquidity)
Price	44	8	BinaryPrice	*Price* must be an applicable minimum trading increment.
PurgeRejectReason	58**	1	Text	Reason for the purge rejection. See Section '4.5 - Order Reason Codes'.

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RestatementReason	n/a	1	Alphanumeric	The reason for this Order Restated message. `W` = Would have self-traded
RiskGroupID	7699	2	Binary	Used to group orders for use in order purges where multiple orders can be cancelled by specifying a list of *RiskGroupID*s.
RiskGroupIDCnt	7698	1	Binary	Number of repeating *RiskGroupID* included in this message. Value 0 to 10.
SecondaryOrderID	198	8	Binary	Present on a Self-Trade Protection triggered restatement. Cboe *OrderID* of contra side of prevented match.
Side	54	1	Alphanumeric	`1` = Buy `2` = Sell `5` = Sell Short `6` = Sell Short Exempt
SourceMatchingUnit	-	1	Binary	The matching unit reporting the purge.
Symbol	55	8	Alphanumeric	Instrument symbol (QUICK symbology)
Text	58	60	Text	Human readable text with more information.
TimeInForce	59	1	Alphanumeric	`0` = DAY (default) `3` = IOC (portion not filled immediately is cancelled). `4` = FOK (an IOC where the entire size must be filled immediately, otherwise the order will be cancelled back). `6` = GTD (expires at earlier of specified *ExpireTime* or end of day).
TradingFirm	49	4	Alphanumeric	The four-character identifier of the trading firm.
TransactionTime	60	8	DateTime	The time the event occurred in the Matching Engine (not the time the message was sent).

** In FIX, the first character of Tag 58 (Text) provides the reason code related to the subsequent human-readable text.

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4.5 Order Reason Codes

The following is a list of all reason codes used by Cboe Japan. These reason codes are used in a variety of contexts (order cancellations and order rejections). All reasons are not valid in all contexts. The reason code will be followed by free-form text. The specific text the system delivers may vary from the text listed below to provide clarification of the reject reason. Cboe Japan may add additional reason codes without notice. Participants must gracefully ignore unknown values.

`A` = Admin
- Most rejects due to field validation errors will be reported with a reason of Admin.

`D` = Duplicate identifier (e.g., ClOrdID)
- Duplicate *ClOrdID* detected. Note that Cboe Japan will not detect every possible occurrence of a duplicate by a participant.

`H` = Halted

`I` = Incorrect data center
- Message directed to the DR data center

`K` = Order rate threshold exceeded

`N` = Ran out of liquidity to execute against
- Cancel reason for the unexecuted size of an IOC/FOK order.

`O` = *ClOrdID* does not match a known order

`U` = User requested

`W` = Add liquidity only order would remove

`X` = Order expired
- Cancellation of an expired GTD order

`Y` = Symbol not supported

`Z` = Unforeseen reason

`o` = Max open orders count exceeded

`y` = Order received by Cboe Japan during replay

`z` = Session end

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5. Port Attributes

Attribute	Default	Description
HST code	(none)	Identifies the high speed trader using this port.
Designated as DLP	No	Applicable only to the Select market. "No" – all order must be IOC or FOK "Yes" – all orders must be Post Only
Default RiskGroupID	No	Default value for *RiskGroupID*.
Allow Test Symbols Only	No	Allow or disallow orders in non-test symbols.
Maximum Notional Value Per Order	No	Maximum notional value allowed on any individual order (acceptable value 1 - 10,000,000,000).
Maximum Order Lot Size	9,999,999	Maximum number of lots allowed per order.
Cancel on ME Disconnect	Yes	When set to "No", this setting allows orders to remain open on a Matching Unit failover. When set to "Yes", all open orders associated with a session are immediately cancelled in the event of loss of connectivity to a Matching Unit. If a failover takes longer than five minutes, then all orders are cancelled unconditionally.
Cancel on Disconnect	*Option #1*	Cboe Japan will offer Participants the following options for cancelling orders because of a session disconnect: 1. Cancel All Open Orders (continuous book) 2. Do Not Cancel Any Open Orders
Default *NoTradeFeat*	N (cancel newest)	Specifies default value for *NoTradeFeat* (7713).
Cancel on Reject	Yes	Cancels an order upon a modify reject for that order.
Cancel on Halt	No	Cancel open orders for a symbol upon a halt.
Reject Orders on Drop Port Disconnect	No	Allows a participant to associate one or more drop port(s) to an order entry port(s). Once the association has been established, if no drop ports associated with an order entry port are connected, reject orders on the order entry port until at least one of the drop port session connections have been established.
Reject Orders on DROP Port Timeout (s)	30	Only applicable for sessions where "Reject Orders on Drop Port Disconnect" has been enabled. When the last associated drop port for the order entry session has disconnected, the reject/cancel actions will be taken on the order entry session if an associated drop port has not re-established its connection in the defined time. Minimum value allowed is 20.
Cancel Open Orders on Drop Port Disconnect	No	Only applicable for sessions where "Reject Orders on Drop Port Disconnect" has been enabled. If all drop ports associated with an order entry port become disconnected, cancel all open orders on the order entry port.
Port Order Rate Threshold	Default = 4000 msgs/sec	The maximum allowed message rate on the session. When the first non-administrative message is received,

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	Maximum allowed = 4000 msgs/sec	a one second window begins. During the second no more than 3,999 additional non-administrative messages will be allowed within that window. If the rate is exceeded all new orders in the time window are rejected, modifies are treated as cancels, and cancels are processed.

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6. Support

Please direct questions or comments regarding this specification to TradeDeskJP@cboe.com.

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Appendix A – Login Strategies

Login Message Flow

On a successful login, the following is the sequence of messages to expect. Note that after sending a `Login Request`, participants should wait for both a `Login Response` of "Accepted" as well as a `Replay Complete` message before sending new application messages.

Direction	Message
Participant → Cboe Japan	Login Request
Participant ← Cboe Japan	Login Response (`A` = Accepted)
Participant ← Cboe Japan	For each matching unit, all messages for the participant having sequence number greater than the seq # in the login request for that unit. Possibly no such messages are transmitted, for instance on the first login of the day.
Participant ← Cboe Japan	Replay Complete
Participant ←→ Cboe Japan	(Begin normal flow: Order traffic, or Heartbeats at 1/sec)

Recommended Login

"Request replay of everything that you missed"

The login protocol supports a variety of options for how a participant may want to approach reconciling sequence numbers and messages after a disconnection. For most participants, the following Login strategy is the simplest approach to support a robust, lossless login experience:

Field Name	Value	Comment
StartOfMessage	`B0 E3`	
MessageLength	`1E 00`	30
MessageType	`01 00`	1 (login request)
MatchingUnit	`00`	
Reserved	`00`	
SequenceNumber	`00`	(unsequenced)
SessionId	`41 42 43 44`	(ex. "ABCD") - Session Id as supplied by Cboe Japan
SessionSubId	`30 30 30 31`	(ex. "0001") Session Sub Id as supplied by Cboe Japan

Password	`61 62 63 64`
	`65 66 67 68`
	`69 6A`
		(ex. "abcdefghij") Password

ReplayUnspecifiedUnit	`52`	"R" - Replay unknown units.
NumberOfUnits	`02`	There are 2 rows in the array below, representing the 2 system matching units.

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→*UnitNumber*	01	Matching unit #1
→*UnitSequence*	00 00 00 00	Last *SequenceNumber* that participant received with *MatchUnit*=1. Zero if participant has not received anything for *MatchUnit*=1, for instance as the first login of the day.
→*UnitNumber*	02	Matching unit #2
→*UnitSequence*	00 00 00 00	Last *SequenceNumber* that participant received with *MatchUnit*=1. Zero if participant has not received anything for *MatchUnit*=2, for instance as the first login of the day.
From 3rd August 2025 onwards:		
NumberOfUnits	04	There are 4 rows in the array below, representing the 4 system matching units.
→*UnitNumber*	01	Matching unit #1
→*UnitSequence*	00 00 00 00	Last *SequenceNumber* that participant received with *MatchUnit*=1. Zero if participant has not received anything for *MatchUnit*=1, for instance as the first login of the day.
→*UnitNumber*	02	Matching unit #2
→*UnitSequence*	00 00 00 00	Last *SequenceNumber* that participant received with *MatchUnit*=1. Zero if participant has not received anything for *MatchUnit*=2, for instance as the first login of the day.
→*UnitNumber*	03	Matching unit #3
→*UnitSequence*	00 00 00 00	Last *SequenceNumber* that participant received with *MatchUnit*=1. Zero if participant has not received anything for *MatchUnit*=3, for instance as the first login of the day.
→*UnitNumber*	04	Matching unit #4
→*UnitSequence*	00 00 00 00	Last *SequenceNumber* that participant received with *MatchUnit*=1. Zero if participant has not received anything for *MatchUnit*=4, for instance as the first login of the day.

Other Login Strategies

No Replay

"Just start with the next live event"

If the participant is confident that they are disinterested in any missed messages, they may elect to Skip (S) unspecified units and specify no units. This can be used as an initial login. Differences from the recommended login are in **Bold**.

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Field Name	Value	Comment
StartOfMessage	B0 E3	
MessageLength	1E 00	30
MessageType	01 00	1 (login request)
MatchingUnit	00	
Reserved	00	
SequenceNumber	00	(unsequenced)
SessionId	41 42 43 44	(ex. “ABCD”) - Session Id as supplied by Cboe Japan
SessionSubId	30 30 30 31	(ex. “0001”) Session Sub Id as supplied by Cboe Japan
Password	61 62 63 64 65 66 67 68 69 6A	(ex. “abcdefghij”) Password
ReplayUnspecifiedUnit	**53**	**“S” – Skip replay of unspecified units.**
NumberOfUnits	**00**	**There are no rows in the array below.**

Fail Login if any units are unspecified

A minor variation on the “Recommended Login”, this strategy will fail the login if not all known units are specified. This would provide some protection against a misconfiguration, or the unexpected introduction of additional matching units at Cboe Japan (although Cboe Japan would provide extensive notice and planning before introducing additional units). Differences from the recommended login are in **Bold**.

Field Name	Value	Comment
StartOfMessage	B0 E3	
MessageLength	1E 00	30
MessageType	01 00	1 (login request)
MatchingUnit	00	
Reserved	00	
SequenceNumber	00	(unsequenced)
SessionId	41 42 43 44	(ex. “ABCD”) - Session Id as supplied by Cboe Japan
SessionSubId	30 30 30 31	(ex. “0001”) Session Sub Id as supplied by Cboe Japan
Password	61 62 63 64 65 66 67 68 69 6A	(ex. “abcdefghij”) Password
ReplayUnspecifiedUnit	**46**	**“F” – Fail unknown units.**
NumberOfUnits	02	There are 2 rows in the array below, representing the 2 system matching units.

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→*UnitNumber*	01	Matching unit #1
→*UnitSequence*	00 00 00 00	Last *SequenceNumber* that participant received with *MatchUnit*=1. Zero if participant has not received anything for *MatchUnit*=1, for instance as the first login of the day.
→*UnitNumber*	02	Matching unit #2
→*UnitSequence*	00 00 00 00	Last *SequenceNumber* that participant received with *MatchUnit*=1. Zero if participant has not received anything for *MatchUnit*=2, for instance as the first login of the day.
From 3rd August 2025 onwards:		
NumberOfUnits	04	There are 4 rows in the array below, representing the 4 system matching units.
→*UnitNumber*	01	Matching unit #1
→*UnitSequence*	00 00 00 00	Last *SequenceNumber* that participant received with *MatchUnit*=1. Zero if participant has not received anything for *MatchUnit*=1, for instance as the first login of the day.
→*UnitNumber*	02	Matching unit #2
→*UnitSequence*	00 00 00 00	Last *SequenceNumber* that participant received with *MatchUnit*=1. Zero if participant has not received anything for *MatchUnit*=2, for instance as the first login of the day.
→*UnitNumber*	03	Matching unit #3
→*UnitSequence*	00 00 00 00	Last *SequenceNumber* that participant received with *MatchUnit*=1. Zero if participant has not received anything for *MatchUnit*=3, for instance as the first login of the day.
→*UnitNumber*	04	Matching unit #4
→*UnitSequence*	00 00 00 00	Last *SequenceNumber* that participant received with *MatchUnit*=1. Zero if participant has not received anything for *MatchUnit*=4, for instance as the first login of the day.

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Revision History

Document Version	Date	Description
1.0.0	22/11/23	Initial draft.
1.0.1	22/12/15	Added "default RiskGroupID" port attribute. Updated *MessageLength* in Session Message examples. Added NumberofUnits, UnitNumber, and UnitSequence fields to Example Login Request Messages. Updated field offsets in `New Order` messages. Updated "*MarginTransaction*" to "*MarginClassification*" in `Order Acknowledgement` messages. Updated several datatypes from Text to Alpha/Alphanumeric. Corrected minor typos and capitalization throughout. Removed unused *ExecRefID*, *OrigTime*, *TradingSessionMode*, and *TradingSessionStatus* fields from Section 4. Removed `Trade Cancel` message. Added *OrderID* to `Order Modified` messages. Added Example messages for all Application messages.
1.0.2	23/02/10	Corrected FIX tag cross-reference for *TradingFirm* to be SenderCompID(49).
1.0.3	23/05/01	Updated Cancel on Disconnect port attribute description. Renamed *MarginClassification* field to *MarginTransactionType*.
1.0.4	23/07/18	Updated description of Trading Firm Filter in Section 4.1.4 Purge Orders. Updated "Cancel on ME Disconnect" port attribute default to "Yes".
1.0.5	23/07/25	Corrected example message values in Sections 3.1.1.1 and 3.2.1.1. Replaced "Maximum Order Size" port attribute in Section 5 with "Maximum Order Lot Size". Updated *RiskGroupID* description in `Order Acknowledgement`.
1.0.6	23/07/26	Clarified that the "New Order" *RiskGroupID* port default value will be applied to a zero *RiskGroupID*.
1.0.7	23/08/22	Corrected *DateTime* example in section 1.4: Data Types. Added *Text* (58) reason code 'W'.
1.0.8	23/02/26	Updated section 1.5 to include latency expectations as well as Members' responsibility to monitor the status of the messages they send to the exchange.
1.0.9	24/07/05	Updated Maximum Notional Value Per Order Port Attribute.
1.0.10	15/01/25	Updated with Cboe Titanium branding.
1.0.11	15/05/25	Updated specification title to include "Version 3".

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		Updated Sections 1.5.4.2 and 3.1.1: August 3rd 2025 onwards, Cboe Japan will have 4 matching units for Alpha, and 4 matching units for Select. Updated Appendix A with recommended login and failed logic if unit number unspecified with matching unit increase.
1.0.12	25/06/09	Updated "Maximum Notional Value Per Order" port attribute to 1 - 10,000,000,000.

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